Fractional indices

1 Write the following in surd form:

a $x^{\frac{1}{3}}$ b $a^{\frac{3}{2}}$ c $20^{\frac{5}{4}}$ d $y^{-\frac{1}{2}}$

e $3x^{\frac{1}{2}}$ f $b^{2\frac{1}{3}}$ g $y^{1.5}$ h $5^{-\frac{4}{3}}$

2 Write the following as a single power of 2:

a $\sqrt[3]{2}$ b $\sqrt[5]{16}$ c $\sqrt{32}$ d $2\sqrt{2}$

e $8\sqrt[3]{2}$ f $2^a \times \sqrt{2}$ g $\dfrac{8}{\sqrt{2}}$ h $\dfrac{1}{\sqrt{8}}$

3 Write the following as a single power of 3:

a $\sqrt{27}$ b $3^x \times \sqrt[4]{3}$ c $9\sqrt{3}$ d $\dfrac{1}{\sqrt[3]{9^x}}$

4 Complete the following statement:

$$\begin{aligned}\sqrt{m^8} &= \left(m^8\right)^{\square} \\ &= m^{\square \times \frac{1}{2}} \\ &= m^{\square}\end{aligned}$$

5 Write the following in the form x^k, where k is rational:

a $\sqrt{x}$ b $\sqrt[6]{x}$ c $\dfrac{1}{\sqrt{x}}$ d $\left(\sqrt[3]{x}\right)^2$

e $\sqrt[4]{x^3}$ f $\left(\sqrt[7]{x}\right)^6$ g $\sqrt[4]{x^8}$ h $\dfrac{1}{\sqrt{x}}$

i $\sqrt{\sqrt{x}}$ j $\sqrt[3]{x^{6a}}$ k $\dfrac{1}{\sqrt{x^{-3}}}$ l $x\sqrt{x}$

m $\dfrac{1}{\sqrt[4]{x^5}}$ n $\sqrt[5]{\sqrt{x}}$ o $x^2 \times \sqrt[3]{x}$

6 Consider the expression $64^{\frac{2}{3}}$.

a Complete the statement: $64^{\frac{2}{3}} = \left(\sqrt[\square]{64}\right)^{\square}$ b Hence, evaluate $64^{\frac{2}{3}}$.

7 Without using a calculator, evaluate the following:

- a $1^{\frac{1}{10}}$
- b $4^{\frac{3}{2}}$
- c $121^{\frac{1}{2}}$
- d $\sqrt[3]{27}$
- e $125^{\frac{1}{3}}$
- f $(-64)^{\frac{1}{3}}$
- g $64^{-\frac{1}{6}}$
- h $81^{-\frac{3}{4}}$
- i $\sqrt[3]{-64}$
- j $\left(\frac{9}{100}\right)^{\frac{1}{2}}$
- k $\left(64^{\frac{1}{9}}\right)^{\frac{9}{2}}$
- l $\left(\frac{8}{125}\right)^{\frac{2}{3}}$
- m $(-32)^{\frac{4}{5}}$
- n $\dfrac{64^{\frac{1}{3}}}{64^{\frac{2}{3}}}$
- o $\dfrac{1}{-\sqrt{169}}$
- p $\sqrt[3]{\dfrac{-64}{125}}$
- q $\dfrac{\sqrt[3]{40}}{\sqrt[3]{5}}$
- r $36^{\frac{1}{2}} - 32^{\frac{3}{5}}$
- s $1000^{\frac{8}{9}} \times 1000^{-\frac{5}{9}}$
- t $\sqrt[3]{\sqrt[4]{16} + \sqrt{625}}$

8 Use a calculator to evaluate the following, to two decimal places:

- a $10^{\frac{3}{2}}$
- b $4^{\frac{5}{3}}$
- c $\sqrt[4]{5^3}$
- d $\left(\sqrt[3]{12}\right)^5$

9 Simplify the following expressions, giving your answers in index form. Assume that all variables represent positive numbers.

- a $\left(\sqrt{b}\right)^8$
- b $\sqrt{m^6}$
- c $\sqrt[4]{a^5}$
- d $\dfrac{1}{\sqrt[5]{a^6}}$
- e $\sqrt[3]{x^6}$
- f $\sqrt[3]{m^3}$
- g $\sqrt{j^2x^4}$
- h $\left(\sqrt[4]{x^5y^3}\right)^{24}$
- i $\left(4a^8\right)^{\frac{1}{2}}$
- j $\left(625u^{16}v^{12}\right)^{\frac{1}{2}}$
- k $8b^{\frac{3}{4}} \div 2b^{\frac{2}{3}}$
- l $y^3 \times \sqrt[3]{y}$
- m $\sqrt{(2x+9)^2}$
- n $\sqrt[3]{9^3x^{18}y^{12}}$
- o $\left(\dfrac{x^{15}}{1024}\right)^{\frac{2}{5}}$
- p $\sqrt{\dfrac{36x^{18}}{y^{20}}}$
- q $\sqrt{16y^2 + 24y + 9}$
- r $\sqrt{x^2y^2 + 18xy^2 + 81y^2}$

10 Simplify the following expressions, giving your answers in surd form. Assume that all variables represent positive numbers.

- a $\dfrac{\sqrt{x^2 + 5x + 6}}{\sqrt{x + 2}}$
- b $\dfrac{\sqrt[3]{x^2 + 9x + 20}}{\sqrt[3]{x + 4}}$

11 Describe how we could interpret the expression $m^{\frac{q}{r}}$ in terms of powers and roots of m.

12 Determine whether the following statements accurately describe the meaning of the expression $x^{-\frac{y}{z}}$:

a $x^{-\frac{y}{z}}$ means we are raising x to the power of $\frac{z}{y}$, then taking the reciprocal of the result.

b $x^{-\frac{y}{z}}$ means we are taking the reciprocal of x, then raising the result to the power of $\frac{y}{z}$.

c $x^{-\frac{y}{z}}$ means we are taking the reciprocal of x, then raising the result to the power of $\frac{z}{y}$.

d $x^{-\frac{y}{z}}$ means we are raising x to the power of $\frac{y}{z}$, then taking the reciprocal of the result.

13 Is there a real number that equals $\sqrt[4]{-16}$? Explain your answer.

14 Consider the expression $m^5 \times m\sqrt{m}$.

a Express it in simplest index form.

b Express it in surd form.

15 Consider the expression $m^{\frac{1}{4}} = 8^{\frac{1}{2}}$.

a Complete the following working:

$$\begin{aligned} m^{\frac{1}{4}} &= \left(m^{\square}\right)^{\frac{1}{2}} \\ &= \sqrt{\square}^{\frac{1}{2}} \end{aligned}$$

b Hence, or otherwise, solve the equation $m^{\frac{1}{4}} = 8^{\frac{1}{2}}$.

16 Consider the equation $x^{\frac{2}{3}} = 9$.

a Complete the statement: $\left(x^{\frac{2}{3}}\right)^{\square} = x$

b Hence, solve the equation $x^{\frac{2}{3}} = 9$.

17 Find the missing values in the following equations:

a $16^{\frac{3}{\square}} = 64$

b $(\square)^{\frac{2}{3}} = 25$

c $625^{\frac{3}{\square}} = 125$

d $(\square)^{\frac{4}{3}} = 81$

18 Solve the following equations for x:

a $x^{\frac{3}{4}} = 125$

b $(x-4)^{0.5} = 4$

c $(2x-1)^{\frac{2}{3}} = 9$

d $(4x+3)^{-1.5} = 8$

19 Solve the following equation for k:

$$\sqrt[k]{y} \times \sqrt[k]{y} \times \sqrt[k]{y} = y^{\frac{1}{2}}$$

20 To evaluate $81^{\frac{3}{2}}$ would it be more efficient [redacted] e the property $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$, or the property $a^{\frac{m}{n}} = \sqrt[n]{a^m}$. Explain your answer.

Applications

21 The surface area A of a cube with a volume V is given by $A = 6V^{\frac{2}{3}}$.

a Find the surface area of a cube with a volume of 8 cm^3.

b Find the volume of a cube with a surface area of 216 cm^2.

22 The volume V of a sphere with surface area A is given by:

$$V = \frac{A^{\frac{3}{2}}}{6\sqrt{\pi}}$$

a Find the volume of a sphere with a surface area of 50 cm^2. Round you answer to the nearest cubic centimetre.

b Find the surface area of a sphere with a volume of 125 cm^3. Round you answer to the nearest square centimetre.

23 The volume V of a regular tetrahedron with edge length a is given by:

$$V = \frac{a^3}{6\sqrt{2}}$$

a Find the volume of a tetrahedron with a side length of 2 cm.

b Rearrange the formula to make a the subject.

c Hence, find the side length of a tetrahedon with a volume of 72 cm^3.

24 A company's net profit (in dollars) can be modelled by the equation:

$$P = 50n^{\frac{3}{4}} - 2500$$

Where n is the number of units sold.

a Find its net profit be in dollars if it manages to sell 256 units.

b Find the number of units needed to be sold to make a net profit of $\$8300$.

25 A company's net profit (in dollars) can be modelled by the equation:

$$P = 180n^{\frac{3}{5}} - 4000$$

Where n is the number of units sold.

a Find its net profit be in dollars if it manages to sell 32 units.

b Find the number of units needed to be sold to make a net profit of $\$45\,000$. Round to the nearest whole unit.